

WhatNext Vision Motors

Salesforce CRM Implementation

Customer ordering, dealer assignment and stock automation

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01 Executive Summary

WhatNext Vision Motors set out to improve how customers order vehicles and to reduce manual effort for its staff. The Salesforce solution centralises vehicle, dealer, customer, order, test drive and service data, links each order to a dealer, blocks orders for vehicles that are out of stock, reminds customers of upcoming test drives, and uses a scheduled batch process to keep order statuses aligned with real stock levels.

AT A GLANCE

- 6 custom objects
- 2 record-triggered flows
- 1 trigger with handler class
- 2 batch and scheduler classes

02 Scope and Roadmap

Each requirement area was mapped to a concrete deliverable, then built in seven sequential milestones.

REQUIREMENT	BUSINESS NEED	DELIVERED
Salesforce CRM	Track vehicles, stock, dealers, orders, test drives and service requests	Six custom objects, tabs and a Lightning app
Process automation	Assign a dealer to each order; remind customers about test drives	Auto Assign Dealer and Test Drive Reminder flows
Apex and triggers	Enforce stock rules in maintainable code	VehicleOrderTrigger and VehicleOrderTriggerHandler
Batch jobs	Reconcile pending orders with stock	VehicleOrderBatch run by a scheduler class

MILESTONE 1	MILESTONE 2	MILESTONE 3	MILESTONE 4	MILESTONE 5	MILESTONE 6	MILESTONE 7
Developer org	Objects	Tabs	Lightning app	Fields	Flows	Apex
Org created, verified and secured	Vehicle, Dealer and remaining objects	Tab for each custom object	App and navigation items	Picklists, numbers, dates, lookups	Dealer and reminder flows	Trigger, batch and schedule

Environment Setup

- A Salesforce Developer Edition org was created, the account verified by email, and a password and security question set.
- All configuration was completed from Setup using Object Manager, App Manager, Flow Builder and the Developer Console.

03 Data Model

Vehicle Customer and Vehicle sit at the centre of the model. Orders, test drives and service requests each reference both, and Vehicle references its authorised dealer, so every order can be traced to a dealer through its vehicle.

Vehicle <i>Vehicle details and stock</i> Vehicle Model (picklist) Stock_Quantity__c (number) Price__c (currency) Status__c (picklist) Links: Lookup to Vehicle Dealer	Vehicle Dealer <i>Authorised dealer information</i> Dealer_Name__c, Dealer_Location__c Dealer_Code__c (auto number) Phone__c, Email__c Links: Referenced by Vehicle	Vehicle Customer <i>Customer contact details</i> Customer_Name__c Email__c, Phone__c, Address__c Preferred_Vehicle_Type__c Links: Referenced by orders, test drives, service requests
Vehicle Order <i>Tracks vehicle purchases</i> Order_Date__c (date) Status__c: Pending, Confirmed, Delivered, Canceled Links: Lookups to Customer and Vehicle	Vehicle Test Drive <i>Tracks test drive bookings</i> Test_Drive_Date__c (date) Status__c: Scheduled, Completed, Canceled Links: Lookups to Customer and Vehicle	Vehicle Service Request <i>Tracks servicing requests</i> Service_Date__c, Issue_Description__c Status__c: Requested, In Progress, Completed Links: Lookups to Customer and Vehicle

Picklists use restricted value sets, which keeps status data clean and gives the flows and Apex code reliable values to test against.

04 Application and Interface

COMPONENT	WHAT WAS BUILT	CONFIGURATION
Custom tabs	One tab per custom object	Created from Setup with default profile and app visibility
Lightning app	WhatNext Vision Motors	Built in App Manager with default options and utility items
Navigation items	Vehicle, Dealer, Customer, Order, Test Drive, Service Request, Reports, Dashboard	Custom objects selected from the item list
User access	System Administrator profile	Can be extended to sales and service profiles later

05 Process Automation

Flow 1: Auto Assign Dealer

Runs when a Vehicle Order is created and links the order to a dealer matching the customer's location.



ELEMENT	CONFIGURATION
Start	Object Vehicle Order; trigger on record created; entry conditions use All Conditions Are Met (AND)
Get Customer Information	Vehicle Customer where Id equals {!\$Record.Vehicle_Customer__c}
Get Nearest Dealer	Vehicle Dealer where Dealer_Location__c equals {!Get_Customer_Information.Address__c}
Assign Dealer to Order	Update Records using the record returned by Get Nearest Dealer; saved and activated as Auto Assign Dealer

Flow 2: Test Drive Reminder

Runs when a Vehicle Test Drive is created or updated with Status Scheduled, and emails the customer one day ahead.



ELEMENT	CONFIGURATION
Scheduled path	Reminder Before Test Drive; time source Test_Drive_Date__c; offset 1 Day Before
Get Customer Information	Vehicle Customer by Id, first record only, all fields stored automatically
Text template EmailSent	Rich-text greeting naming the customer and vehicle, signed by the WhatNext Vision Motors team
Send Test Drive Reminder	Send Email action; subject "Reminder: Your Test Drive is Tomorrow!"; recipient is the customer Email__c; rich-text body and line breaks on

A test run confirmed the reminder reached the customer's inbox with a personalised greeting and message body. Both flows were saved with clear labels and activated.

Why Flows

- Administrators can change dealer or reminder logic without writing code.
- Scheduled paths remove the need for a separate job to send test drive reminders.
- Declarative automation sits alongside Apex, so each rule uses the simplest tool that fits.

06 Apex Development

Rules that go beyond declarative tools use the trigger handler pattern. The trigger holds no logic and passes control to a handler class, which keeps the code modular, testable and easy to extend.

Trigger

```
trigger VehicleOrderTrigger on Vehicle_Order__c (before insert, before update,
                                                after insert, after update) {
    VehicleOrderTriggerHandler.handleTrigger(trigger.new, trigger.oldMap,
                                              trigger.isBefore, trigger.isAfter, trigger.isInsert, trigger.isUpdate);
}
```

HANDLER METHOD	WHEN IT RUNS	BEHAVIOUR
preventOrderIfOutOfStock	Before insert and update	Queries Stock_Quantity__c once for all vehicles and calls addError when stock is zero or less
updateStockOnOrderPlacement	After insert and update	For Confirmed orders, reduces vehicle stock by one and updates all vehicles in one DML statement

Both methods are bulk-safe: Ids are gathered in sets, one SOQL query is used, and no query or DML sits inside a loop.

```
if (vehicle.Stock_Quantity__c <= 0) {
    order.addError('This vehicle is out of stock. Order cannot be placed.');
```

Design Principles

- **Separation of concerns.** Trigger routes events; the handler owns the business rules.
- **Bulk safety.** Every method handles lists of records, so data loads and batch updates stay within governor limits.
- **Clear feedback.** Users see a plain-language error on the record when an order cannot be placed.

Batch Job and Scheduler

VehicleOrderBatch selects every order with Status Pending. For each one, if the vehicle has stock, the order becomes Confirmed and stock falls by one. VehicleOrderBatchScheduler runs the batch with a scope of 50 and was scheduled from the Developer Console; the job appears in Setup under Scheduled Jobs as Daily Vehicle Order Processing.

```
SELECT Id, Status__c, Vehicle__c FROM Vehicle_Order__c WHERE Status__c = 'Pending'
```

```
String cronExp = '0 0 12 * * ?';
System.schedule('Daily Vehicle Order Processing', cronExp, new VehicleOrderBatchScheduler());
```

07 Validation and Recommendations

PASSED Out-of-stock order Save blocked with an error message	PASSED Scheduled test drive Reminder email received	PASSED Batch scheduling Job listed in Scheduled Jobs
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Recommendations

The review found points worth hardening before wider production use.

- 01 Nearest dealer logic.** The flow matches Dealer_Location__c exactly to the customer address, so it finds equal text, not the closest dealer. Use Geolocation fields with DISTANCE() in SOQL, or match on city or postal code.
- 02 Single stock decrement.** The trigger reduces stock for each Confirmed order and the batch reduces it again when confirming. Decrement in one place only, and only when status changes to Confirmed.
- 03 Pending orders.** The before-trigger rejects out-of-stock orders, so the Pending orders the batch processes can never be created. Allow them to save as Pending.
- 04 Schedule timing.** The cron expression runs at 12:00 PM while the milestone describes midnight. Use 0 0 0 * * ? for a midnight run.
- 05 Email content.** The reminder shows the vehicle record Id and a support email placeholder. Reference the vehicle name and add a real address.
- 06 Test coverage.** No Apex test classes are included. Add tests for the trigger and batch to meet the 75 percent deployment requirement.

08 Conclusion

The implementation delivers the core of the WhatNext Vision Motors vision: one Salesforce platform that captures vehicle, dealer and customer data, automates dealer assignment and test drive reminders, enforces stock rules in code, and keeps order status current through a scheduled batch process. With the recommendations applied, the solution will be more accurate, easier to maintain and ready to scale.